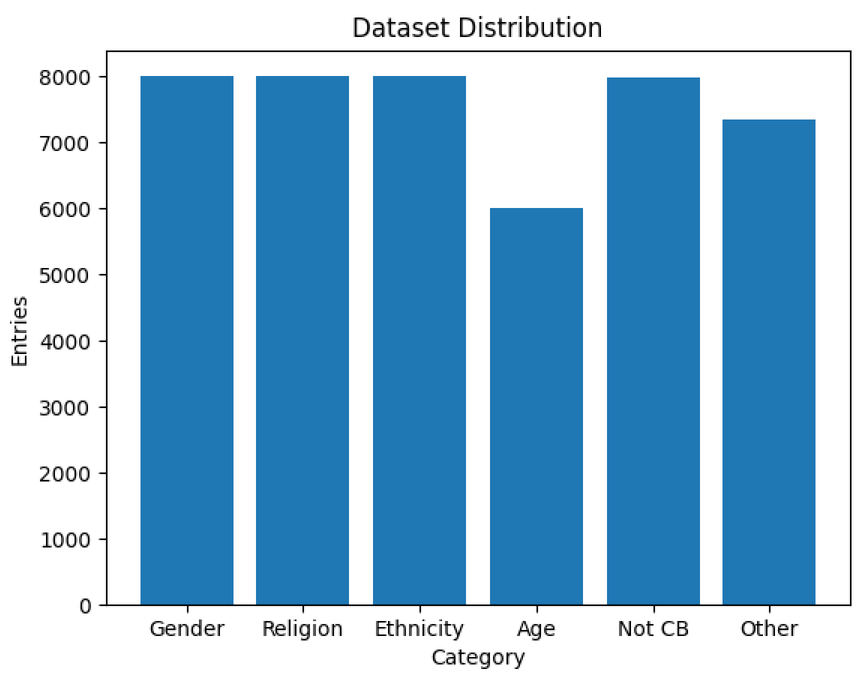
Cyberbullying Detection Dashboard Report

Dataset reference:
https://www.kaggle.com/datasets/andrewmvd/cyberbullyingclassification/data

Project Overview

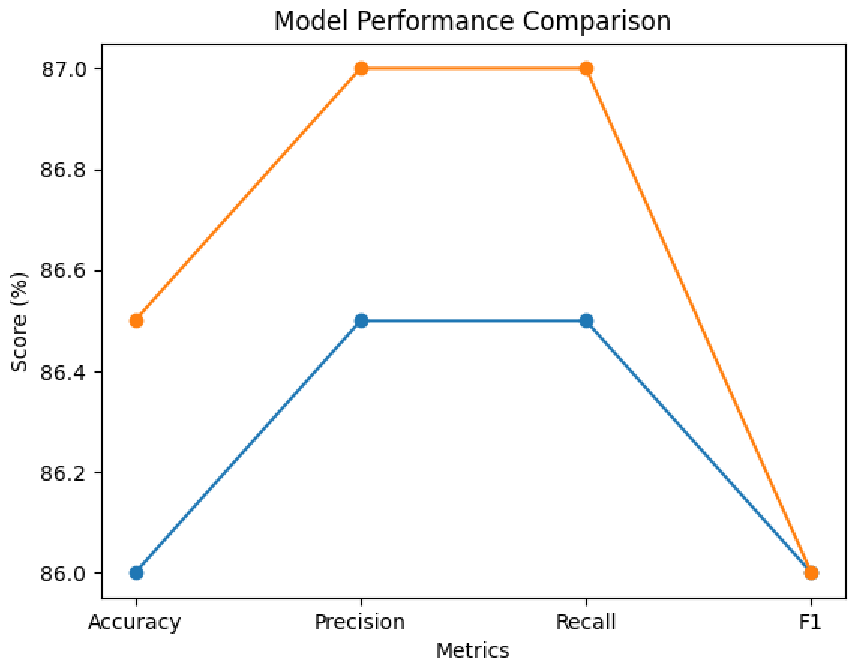
This project focuses on detecting cyberbullying on X (formerly Twitter) using BERT and NLP. It addresses the challenge of identifying harmful content at scale using machine learning.

Dataset Insights



The dataset consists of 46,017 labeled tweets categorized into different types of cyberbullying including gender, religion, ethnicity, and age. The distribution is relatively balanced across categories.

Model Performance



Two training approaches were used: predefined hyperparameters (Blue line) and fine-tuned hyperparameters using GridSearch (Orange line). Both approaches achieved strong performance, with fine-tuning providing a slight improvement.

Key Metrics Summary

Metric	Value
Best Accuracy	87%
Best Precision	87%
Best Recall	87%
Best F1-Score	86%

Insights & Conclusion

- BERT shows strong and consistent performance (>85%).
- Fine-tuning improves results slightly (~1%).
- The model is effective even with limited training data.

Future improvements include scaling the dataset, using more powerful hardware, and deploying a real-time system.